

Finance Analytics Project

How to explain it in an interview

30-second pitch

“I built an end-to-end finance analytics project on about 50,000 banking transactions. I cleaned and validated the data in Python, wrote the MySQL analysis layer for the KPIs and trends, and designed the Power BI dashboard requirements. The dashboard tracks total amount, transaction volume, fees, tax, YoY growth, and performance by month, state, customer segment, transaction type, and gender. It lets management see where revenue comes from and where transactions fail.”

Structure for the longer answer

1. Business problem

A financial organisation couldn't easily track transaction growth, success vs failure, segment and state contribution, or fees and tax in one place. They wanted an interactive dashboard with filters for year, occupation, and category, plus a detail grid for drill-down.

2. Data and cleaning (Python)

- 50,001 rows and 41 columns, covering 2023 to 2026.
- I standardised currency values (INR, inr, inR), derived year, month, quarter, age group, and income segment, and created flags for success, fraud, and outliers.
- I built the risk categories and a net transaction value.

3. SQL analysis (MySQL)

- Data quality checks first: nulls, duplicates, negative amounts, and future-dated records.
- KPI queries with YoY growth using LAG().
- Window functions for rankings, running totals, month-over-month growth, and top-N per group.
- Each dashboard chart has a matching query, so I could cross-check Power BI numbers against SQL.

4. Dashboard design (Power BI)

- Five KPI cards with YoY comparison.
- A monthly line/area chart for seasonality.
- Donut charts for status and gender.
- Horizontal bars for segment and state.
- A matrix for transaction type (amount, fees, tax, count).
- A second page with a detail grid and drill-down.

Findings to mention

Quote real numbers, since they make you credible:

- Total transaction value is about ₹455M across 50,001 transactions, with an average of about ₹9,108.
- Maharashtra is the top state at about 15% of value, followed by Karnataka and Gujarat.
- Loan EMI and Transfer are the largest transaction types by value.
- The success rate is about 85.7% (42,870 of 50,001 transactions). The rest are failed (5,096) or pending (2,035).
- The fraud rate is about 1.3% (630 transactions).
- Monthly volumes are fairly stable, with February consistently the lowest month.

Challenges (interviewers love this part)

- **Hidden data bug:** fee_clean, tax_clean, and total_charges were all zero, so any report built on them would have shown zero fees. I caught it during validation and rebuilt them from the raw fee_amount and tax_amount columns. This shows you validate data rather than trusting it.
- **Misleading YoY:** 2026 shows about -67% growth because it's only a partial year. I flagged it and recommended comparing like-for-like periods.
- **Data quality issues:** I found 326 future-dated records, 8 negative amounts, 1 duplicate transaction ID, and 24 null fees. I handled each with COALESCE or documented it.
- **Missing age data:** 43% of rows had no age. I kept them as an "Unknown" group rather than dropping them, which would have lost almost half the value.

Likely follow-up questions

- **"Why SQL and not just Power BI?"** I use SQL to validate numbers independently and to handle heavy aggregation before visualisation.
- **"How did you calculate YoY?"** I used LAG() over yearly totals: (current - previous) / previous × 100.
- **"What would you do next?"** Build a fraud-risk model, add a forecast for monthly volume, and set up alerts for failed-transaction spikes.
- **"What business recommendation came out of it?"** Focus on the top states and segments, investigate failed transactions in the channels where they cluster, and review fee structure by transaction type.

Tips

- Practice saying the pitch out loud in under a minute.
- Be ready to explain any query on your resume, such as window functions and CTEs.
- Be honest about what was provided (the requirements document) and what you built.